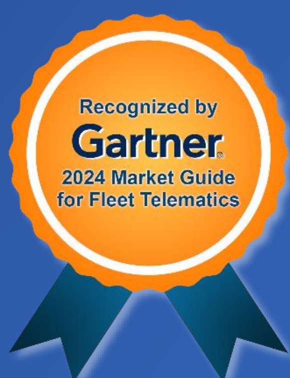


Response to ABG Trading

Technical Proposal Document

Submitted on – December 3, 2024

Corporate HQ
Axestrack Software Solutions Pvt. Ltd.,
Office No.-19, 6th Floor, Jaipur Electronic Market, Near Riddhi Siddhi,
Gopalpura bypass, Jaipur – 302018, Rajasthan, India

Table of Contents

1. **Respondent Information** **3**
1.1 Axestrack Overview 3
1.2 Axestrack Expertise & Recognitions 4
1.2.1 Going beyond real-time visibility (unified platform for logistics) 4
1.2.2 Recognized by analysts 5
1.2.3 Stability with scale 6
2 **Project Overview** **7**
2.1 Our Understanding of ABG Trading Business Process 7
2.2 Scope of Services 8
2.3 Technical Scope 9
2.4 Out of Scope 9
3 **Technical Specification** **10**
3.1 Connected Port Logistics (CPL) Platform Overview 10
3.2 Unique Methodology 11
3.3 To be Process Flow 12
3.4 Stagewise Actions at Checkpoint 12
3.5 To be Digital Layer 14
3.5.1 Vessel Discharge Process 14
3.5.2 Stock Yard loading (Point-of-Sale) Process 14
3.6 Tablet (industrial grade) specification 15
4 **Project Timeline** **17**
4.1 Big Bang Approach Plan 17
4.2 Detailed Project Plan 18
5 **Budget Estimate** **19**
5.1 Cost Estimates 19
5.2 Payment Terms 19
6 **Previous Experience** **20**
7 **Support and Maintenance Process** **23**

8 Governance Structure 24

9 Training Approach 25

10 Key Assumptions & Dependencies 26

11 Overview of Connected Port Logistics 27

11.1 One Stop Solution for Port logistics & Assets Visibility 27
11.2 In-Port Trip Management 27
11.3 SMART Dock & Parking Management 28
11.4 Checklist & Alerts Dashboard 28
11.5 Benefits of Axestrack Connected Port Logistics 29

12 Pragmatic View of Product Roadmap & Innovations Planned 30

1. Respondent Information

1.1 Axestrack Overview

Since its inception in 2008, Axestrack has been at the forefront of transforming the logistics and transportation industry through innovative digital solutions. What began as a small-scale operation in Jaipur has evolved into a leading player in India's logistics landscape.

Our growth story is a testament to the power of entrepreneurship. A team of like-minded individuals united to create a company that would redefine the way logistics is done. Today, Axestrack is proud to serve over 250,000+ commercial vehicles across 100+ cities, offering a comprehensive range of digital solutions. (For more about the platform please visit - https://www.axestrack.com/)

Our flagship product, the Axestrack Operating System (commonly known as aOS), leverages cutting-edge AI technology to optimize supply chain operations. By providing real-time data, analytics, and insights, we empower businesses to make informed decisions, improve efficiency, and reduce cost.

Beyond telematics, we offer a comprehensive suite of digital solutions tailored to meet the unique needs of our clients. From seamless integration with ERP systems to advanced analytics and AI-driven insights, we empower businesses to build a robust digital thread that drives efficiency and growth. By digitizing information and processes, we help our clients stay ahead of the curve and achieve their long-term goals.

At Axestrack, we believe in partnering with our clients to achieve their strategic objectives. By understanding their unique needs, we tailor our solutions to deliver maximum value. Our commitment to innovation, quality, and customer satisfaction has made us a trusted partner for businesses across India.

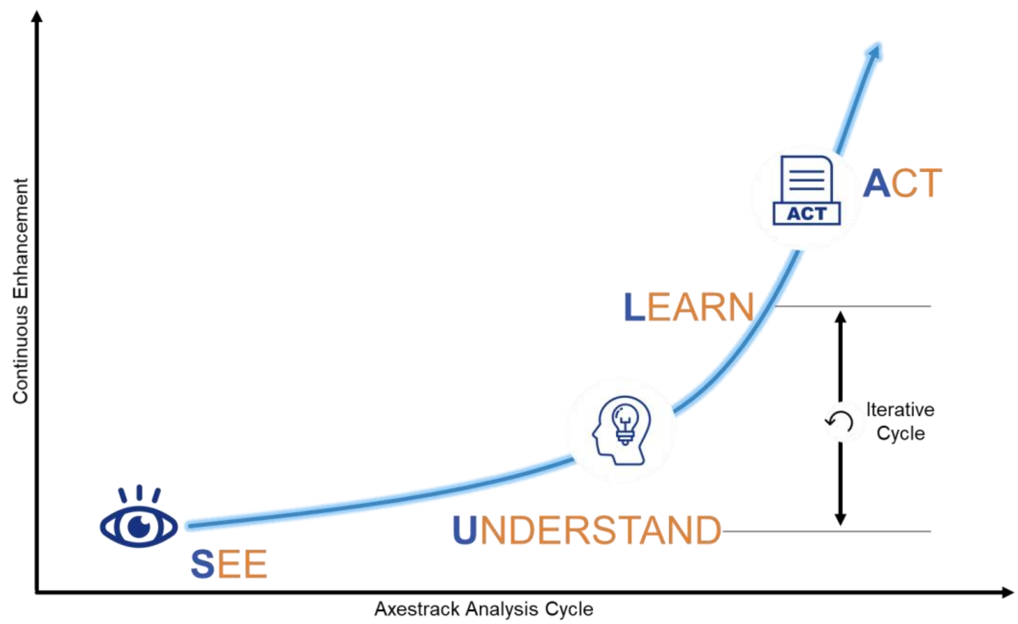
Further, we have accomplished ourselves as true partners to some of the marquee clients in the industry. Key mentions are: -

1.2 Axestrack Expertise & Recognitions

1.2.1 Going beyond real-time visibility (unified platform for logistics)

Axestrack with its suite of products (xSwift, Connected Port Logistics, Managed Control Tower, Connected Plant Logistics, etc.) provides state-of-the-art tracking and visibility to shippers and fleet owners.

The unified, single-pane view across all platforms facilitates rapid and informed decision-making. This connected operating system is structured into three key layers: **See, Understand & Learn**, and **Act**.



- The **See** layer aggregates data from multiple sources, including third-party GPS/Telematics systems, 3PL providers, ERP/TMS platforms, toll systems, government portals, OEM systems, third-party SIMs, and open APIs. This data is consolidated into a comprehensive data pool via a robust integration layer.
- The **Understand & Learn** layer leverages advanced AI and machine learning capabilities to transform this data lake into a data pool, enabling deeper insights and meaningful analytics.
- The **Act** layer presents this processed data through unified platforms, offering actionable analytics and reports. These insights empower the Managed Control Tower to make real-time interventions, ensuring efficient operations and improved outcomes.

1.2.2 Recognized by analysts

Axestrack has been widely recognized as a top digital solution provider through industry research, awards, and strategic partnerships. Axestrack has received numerous awards and accolades for its innovative digital solutions, including Gartner's 2024 Market Guide for Fleet Telematics.

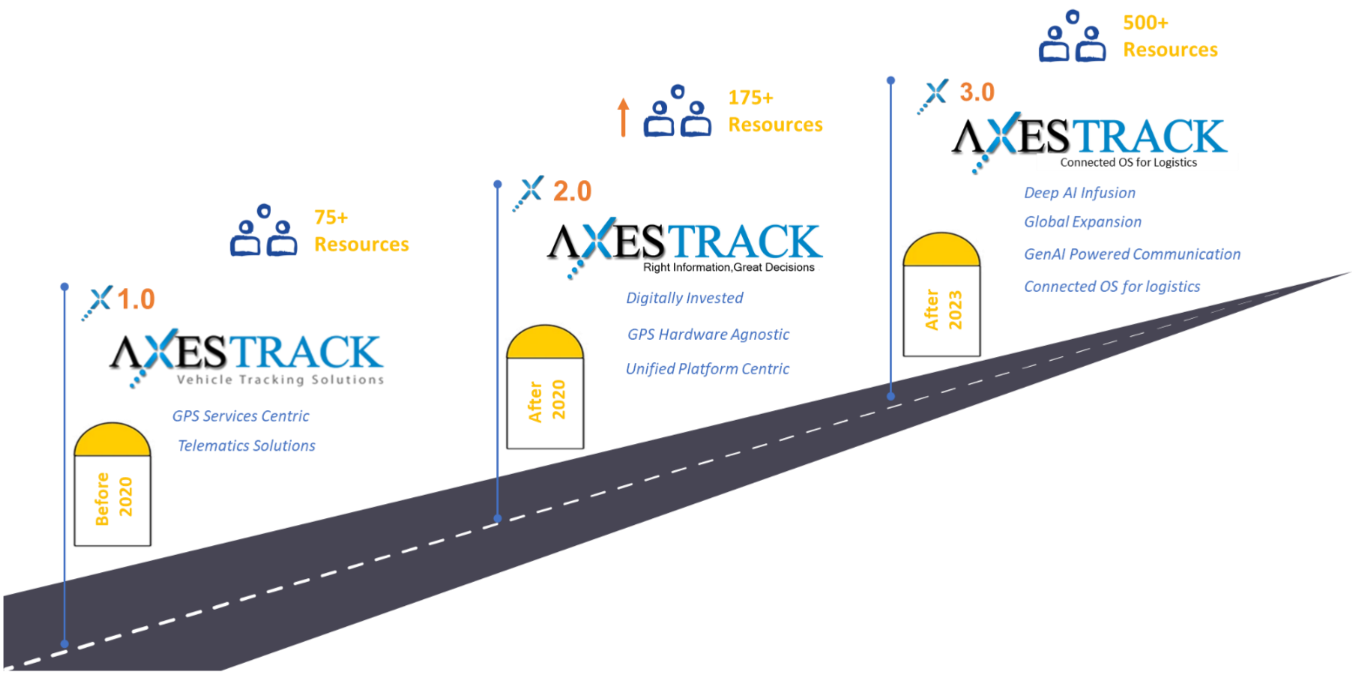
Axestrack believes in excellence which led our innovations and product capabilities recognized by multiple analysts and market leaders to solve transportation challenges in the industry.

According to Gartner 'The fleet telematics market is growing, driven by the increasing demand for efficient fleet management solutions and operational optimization. This growth is catalyzed by widespread telematics adoption, supportive regulatory frameworks, reduced operational costs and enhanced customer satisfaction through real-time visibility on deliveries, driver safety and retention. The integration of AI, ML, and big data analytics is poised to propel the market forward by delivering advanced insights and predictive capabilitiames for fleet management.'

1.2.3 Stability with scale

At Axestrack, we are more than a solution provider—we are strategic partner. With over 15 years of experience in logistics and technology, and the management of our own fleet of 200+ vehicles, we leverage our deep industry expertise to proactively analyze key business KPIs and recommend targeted solutions that address operational gaps. As a bootstrapped and profitable organization, we currently manage 2.5 lakh+ fleets and oversee lakhs of trips monthly.

Our in-house Center of Excellence (CoE) designs, develops, and deploys cutting-edge digital solutions tailored to the logistics industry. These solutions support every stage from proof of concept (POC) to full implementation and enhancement. Our unified platform (**Connected Port Logistics**) seamlessly integrates with any ERP system as a bolt-on solution, enabling faster, more efficient port operations and driving impactful improvements across internal port logistics processes.



2 Project Overview

2.1 Our Understanding of ABG Trading Business Process

ABG Trading faces significant challenges in its port operations, including heavy reliance on manual processes, underutilization of modern technology, and a lack of standardized workflows. These issues result in inefficiencies, errors, and limited visibility across the supply chain. Manual documentation of weighment records, gate passes, and delivery challans creates delays, while the absence of real-time integration with SAP leads to inaccuracies in invoicing and shipment tracking. Additionally, inconsistencies in processes from vessel discharge to the point of sale, coupled with pilferage risks and insufficient compliance mechanisms, further complicate operations.

To address these challenges, Axestrack proposes implementing its flagship solution, Connected Port Logistics (CPL). CPL is a unified digital platform designed to streamline port logistics by automating key touchpoints, digitizing documentation, and integrating real-time data into SAP. The solution includes features such as automated token generation, weighment slip management, and delivery challans, reducing reliance on manual inputs and paperwork. It enhances operational transparency by creating a comprehensive digital audit trail and leverages analytics for process optimization.

CPL will standardize workflows across stages, improve compliance with digital validation checkpoints, and enable real-time tracking of coal shipments. This not only minimizes pilferage risks but also ensures faster and more accurate invoicing. By integrating seamlessly with SAP, CPL provides ABG Trading with a scalable, efficient, and reliable solution, enabling measurable improvements in staff productivity and operational efficiency. This transformation will position ABG Trading as a leader in adopting advanced digital solutions for port operations.

The implementation of Axestrack's CPL will directly address ABG Trading's technological, process-related, and people-centric challenges. It will digitize and streamline the entire port-to-point-of-sale workflow, ensure real-time data integration with SAP, and establish a secure, efficient, and auditable system for managing coal shipments. This alignment ensures long-term operational excellence and a sustainable competitive advantage for ABG Trading.

2.2 Scope of Services

CPL will address the core requirements of ABG Trading by providing the following functionalities:

- Vessel Discharge Flow
    - Capture vessel discharge details, including discharges at berth or mid-sea locations.
    - Record barge/truck details with timestamps and photographic evidence at key stages:
        - When trucks load goods at the berth/jetty.
        - When trucks unload goods at the designated plot.
    - Allow users to upload discharge summary pictures for better accountability.

- Delivery Order Flow
    - Enable recording and management of delivery orders at the time of sale.

- Digital Token Management
    - Assign a unique digital token for each truck.
    - Include details such as truck ID, DO reference, customer reference, time of arrival, and other relevant information.

- Weighment Slip Management
    - Record weight data for each truck post-weighment at the bridge.
    - Generate digital weighment slips that can be printed or emailed.

- Gate Pass Management
    - Capture gate pass details created by the CHA, including truck and weighment data.
    - Send these details to SAP for further processing.

- Delivery Challan Creation
    - Automatically generate delivery challans based on weighment and token data.
    - Include key information like DC number, date, vehicle number, transport document, weight, price, and amount.
    - Send delivery challan details to SAP for seamless integration.

- Assigning User as per Segregation of Duties (Roles and Permissions)
    - Port Operators: Capture and manage tokens, weighment slips, gate pass records, and delivery challans. Access restricted to data specific to their assigned port.
    - Admin Users: Manage user groups, master data, and transactional data.
    - Business Users: Access SAP for invoice management and the application for real-time discharge and delivery data.

- Executing CPL Development and Deployment
- Executing System Integration Testing (SIT) and facilitate User Acceptance Testing (UAT)
- Post Go-live Hypercare Support (1 Month)
- Key User Training through Train the Trainer approach

2.3 Technical Scope

- Seamless Integration with SAP
  - Establish a real-time interface for bidirectional data exchange
    - Delivery order details from SAP to the application.
    - Gate pass and delivery challan details from the application to SAP.

- Build integrations for data to move from Source to end (SAP to CPL; CPL to SAP)
  - Integrations for internal & external reporting
  - Total API Integrations – less than 5j

2.4 Out of Scope

- SAP/ Third Party products performance optimization
- Data cleansing and data extraction from source system
- Issues and bugs related to 3rd Party product or services or plugins
- Customization related to 3rd party products
- Civil work, Mounting poles, Power supply, LAN connectivity, hardware setup, if any.
- Coordination with CHA/Third Party Agent

3 Technical Specification

CPL is a **low-code, no-code** unified platform designed to streamline port operations.

3.1 Connected Port Logistics (CPL) Platform Overview

- **Backend Development:**
  - **Framework:** Spring Framework – Scalable and secure modular architecture.
  - **Database:** PostgreSQL – Feature-rich, structured data handling with high reliability.
  - **Architecture:** Microservices – Independent services to enhance scalability and maintainability.

- **Mobile Application Development (IOS/Android):**
  - **Framework:** Flutter – High-performance cross-platform compatibility for iOS and Android.
  - **Programming Language:** Dart – Optimized for mobile app efficiency and responsiveness.

- **Development Methodologies:**
  - **Agile Development:**
    - Scrum: Iterative sprints with continuous collaboration and feedback.
    - Kanban: Visual task management for optimized workflows.
  - **Design-Driven Development:**
    - User-centric design with tools like Figma/Adobe XD and rigorous user testing during prototyping.

- **DevOps Practices:**
  - **Automation:** CI/CD pipelines with Jenkins for reliable builds, tests, and deployments.
  - **Monitoring:** Integrated logging and performance monitoring

- **Code Versioning and Collaboration:**
  - **Tool:** GitLab – Efficient source control with Git Flow branching strategies.
  - **Collaboration:** Streamlined team workflows for consistency and quality.

- **Quality Assurance:**
  - **Tool:** SonarQube – Continuous code analysis to maintain quality and security standards.
  - **Process:** Regular reviews and fixes for bugs and vulnerabilities.

- **API-First Approach:**
  - API design using Postman for seamless backend and mobile integration.

- **Performance Monitoring:**
  - **Datadog:** Real-time backend performance monitoring and infrastructure health.
  - **Firebase Crashlytics:** Mobile app crash tracking for a smooth user experience.

3.2 Unique Methodology

- **Agile-Driven Development:**
  - Iterative updates aligned with client priorities, ensuring adaptability to evolving needs.
  - User feedback loops integrated into every sprint for tailored solutions.

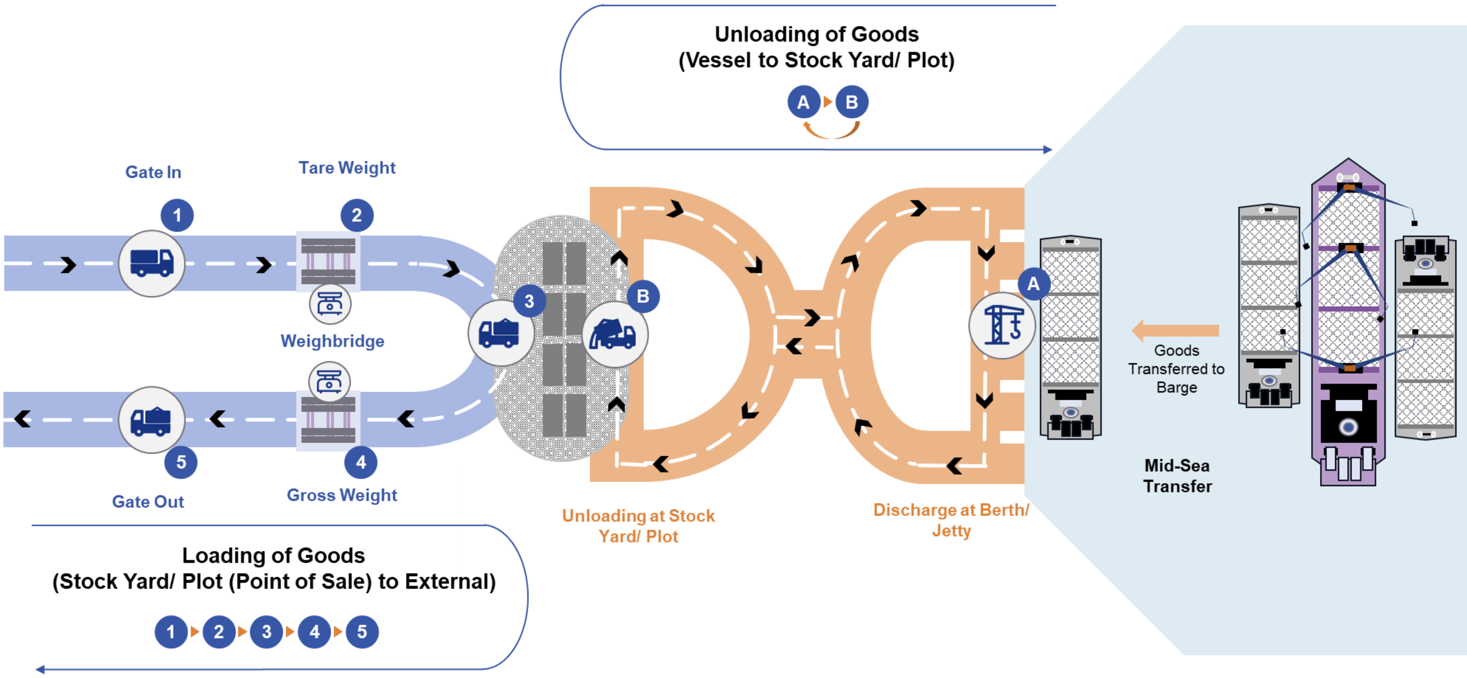
- **Microservices Flexibility:**
  - Services can be independently developed, deployed, and scaled.
  - Fault isolation enhances overall system resilience.

- **API-First Integration:**
  - APIs designed and tested before development to ensure interoperability.
  - Enhances modularity between frontend and backend components.

- **Design-First Philosophy:**
  - Early focus on UX/UI ensures the platform aligns with industry expectations and usability standards.

3.3 To be Process Flow

The diagram below illustrates the overall flow progression through various checkpoints for ABG Trading



3.4 Stagewise Actions at Checkpoint

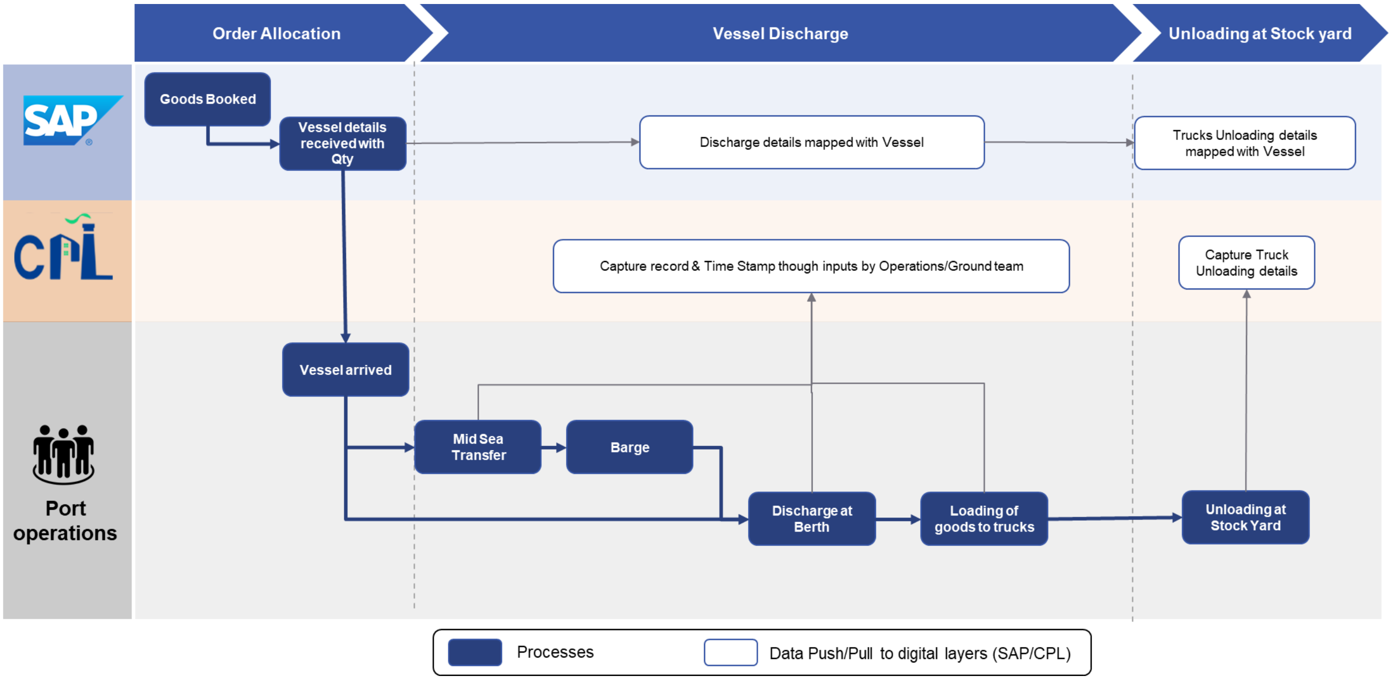
Below checklist will be implemented for all the checkpoints in scope.

Checkpoint	Stagewise Actions
Discharge at Berth/ Jetty	• Vessel & Barge detail • Truck Nos. validation • Image of Discharge • TAT measurement (discharge from vessel, loading to truck) • Image of Loaded Truck • Dynamic Checklist
Unloading at Stock Yard/ Plot	• Truck Nos. validation • Unloading Image of Truck (loaded/unloaded) • TAT measurement • Dynamic Checklist
Mid-Sea Transfer	• Transfer Time • Vessel & Barge detail Note – Subject to input available

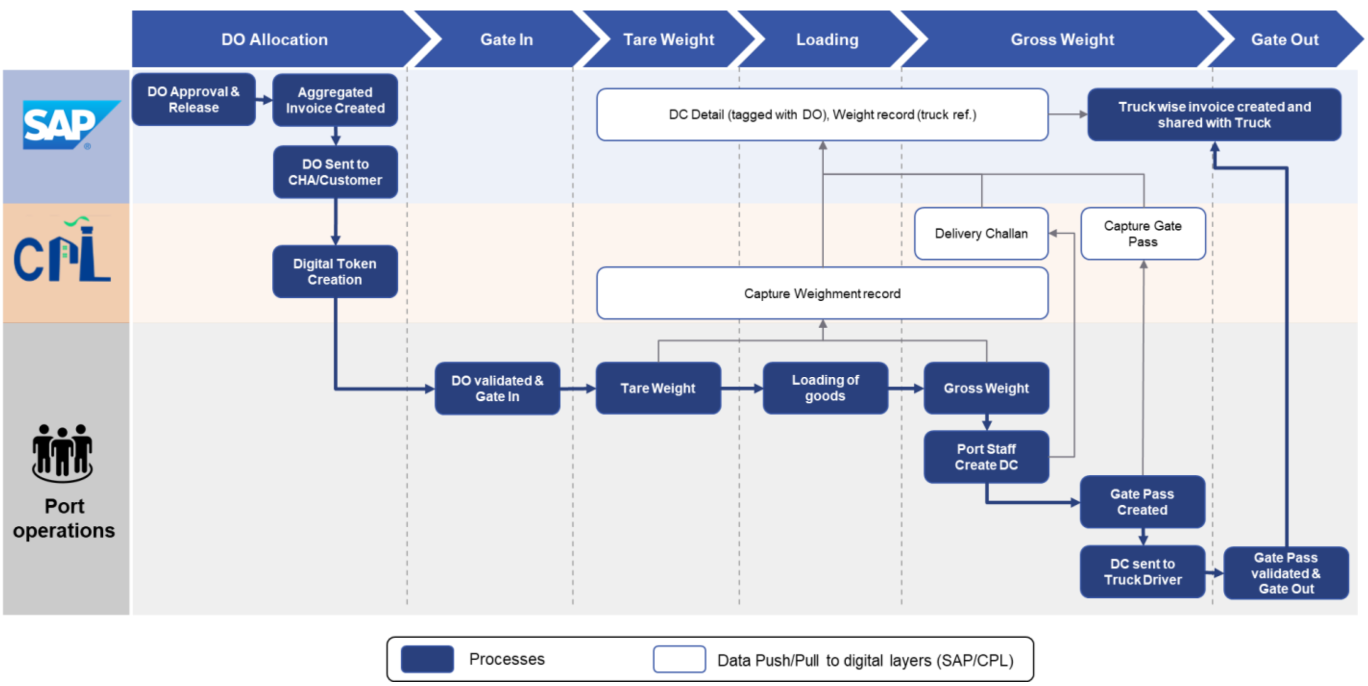
Gate In	• DO verification • Digital Token verification • Image capture • TAT measurement • Dynamic Checklist
Tare Weight	• Tare Weight • DO verification • Token verification • Image capture • TAT measurement • Dynamic Checklist
Loading at Stock Yard/ Plot	• Vehicle/Driver verification • Capture Image of Truck (loaded/unloaded) • DO Check • TAT measurement • Loading Image • Dynamic Checklist
Gross Weight	• Gross/Net Weight • DC upload • Image capture • Weighbridge slip image upload • Dynamic Checklist
Gate Out	• Image capture • Loaded/Unloaded Status • TAT measurement • Dynamic Checklist

3.5 To be Digital Layer

3.5.1 Vessel Discharge Process



3.5.2 Stock Yard loading (Point-of-Sale) Process



3.6 Tablet (industrial grade) specification

We propose to use industrial grade Tablets for data capture, inputs and validation, with below specification.

Details	Specification
Developing Environment	
Operating System	Android 13; GMS&AER, 90-day security updates, FOTA, Soti MobiControl, SafeUEM supported
Performance	
CPU	Octa-core, 2.0GHz; GMS&AER
RAM+ROM	4 GB + 64 GB
User Environment	
Operating Temp.	- 4 °F to 122 °F / -20 °C to 50 °C
Storage Temp.	- 40 °F to 158 °F / -40 °C to 70 °C
Humidity	5% RH - 95% RH non condensing
Drop Specification	Multiple 1.5 m / 4.92 ft. drops to concrete across the operating temperature range
Tumble Specification	500 x 0.5 m / 1.64 ft falls at room temperature
Sealing	IP65 per IEC sealing specifications
ESD	±15 KV air discharge, ±8 KV conductive discharge
Camera	
Front Camera	5 MP
Rear Camera	13 MP Autofocus with flash
Communication	
WLAN	**Support:** 802.11 a/b/g/n/ac/d/e/h/i/k/r/v, 2.4G/5G dual-band, IPV4, IPV6; **Fast roaming:** PMKID caching, 802.11r, OKC; **Operating Channels:** 2.4G(channel 1~13), 5G (channel 36,40,44,48,52,56,60,64,100,104,108,112,116,120,124,128, 132,136,140,144,149,153,157,161,165), Depends on local regulations; **Security and Encryption:** WEP, WPA/WPA2-PSK (TKIP and AES), WAPIPSK-EAP-TTLS, EAP-TLS, PEAP-MSCHAPv2, PEAP-LTS, PEAPGTC, etc.
WWAN	**2G:** 850/900/1800/1900MHz, GPRS, EDGE **3G:** CDMA EVDO: BC0; WCDMA: B1/B2/B5/B8 **4G:** TDD-LTE: B34/B38/B39/B40/B41 FDD-LTE: B1/B2/B3/B4/B5/B7/B8/B12/B13/B17/B20/B28
Bluetooth	Bluetooth 5.1

Vo-LTE	Support Vo-LTE HD video voice call
GNSS	GPS/AGPS, GLONASS, BeiDou, Galileo, internal antenna
Physical Characteristics	
Dimensions	250.8 x 152.0 x 15.0 mm / 9.87 x 5.98 x 0.59 in. (for standard version)
Weight	680 g / 23.99 oz. (for standard version)
Display	8" IPS LTPS 1920 x 1200
Touch Panel	Corning Gorilla Glass, multi-touch panel, gloves and wet hands supported
Power	**Main battery:** Li-ion, rechargeable, 8700 mAh **Standby:** over 500 hours **Continuous use:** over 10 hours (depending on user environment) **Charging time:** 5-7 hours (with standard AC adaptor and USB cable)
Interfaces	USB 2.0 Type-C, OTG
Audio	1 microphone; 1 speaker

4 Project Timeline

4.1 Big Bang Approach Plan

Axestrack proposes a big bang implementation approach, as per below phases

Duration	Activities
W1	• Project Kick Off
W1-W2	• Integration Details • Solution Design, API Developments & Configuration • Unit Testing • Development & Rollout of Checklist
W3	• SIT • Key User Training
W4	• Facilitate UAT • Cutover • Go-Live
W5-W8	• Hypercare & Support

Post Implementation key services include:

- Technical Support: 24/7 help desk and remote diagnostics.
- Maintenance and Updates: Regular software updates.
- Data Management: Continuous data monitoring and analytics.
- Training and Support: User training sessions and resource materials.
- Customization and Optimization: System customization and performance assessments.
- Usability: Provide an intuitive and user-friendly interface for seamless operation.
- Compliance and Security: Ensuring regulatory compliance, robust data protection, secure data exchange, user authentication and enhancing data security.
- Feedback and Improvement: Collecting user feedback and implementing system enhancements.
- Integration Services: Seamless integration with other systems and API support.
- Reporting and Analytics: Custom reports and advanced analytics services.
- Proactive Monitoring: Regular health checks and alerting services.

4.2 Detailed Project Plan

#	Phases	Activities	M1				M2			
			W1	W2	W3	W4	W5	W6	W7	W8
1	**Project Kick Off**	Resources Alignment with ABG Trading								
2	**Solution Discussion**	Solution Workshops (SAP Integration Details)								
3	**Solution Development**	API Developments & Configuration								
4		Unit Testing								
5		Development Sign-off								
6	**Solution Testing**	SIT & SIT Remediation								
8		Key User Training								
9		UAT & UAT Remediation								
11		Testing Sign-off								
12	**Solution Deployment & Support**	DEV to PROD Cutover								
13		Go-Live								
14		Hypercare & Support								

5 Budget Estimate

5.1 Cost Estimates

Please find the detailed cost estimates in the attached PDF document, which accompanies this proposal. We kindly request you to review the budget details provided for a comprehensive understanding of the financial aspects of the project.

5.2 Payment Terms

- Payment will be on milestone basis
  - **Milestone 1:** 100% of implementation cost at the time of project kick-off.
- Post Go-live, the billing will be in advance for 1 year on pro-rata basis.
- 100% payments to be made within 30 days from the date of receipt of original or digitally signed invoice. In case of further delay, overdue charges of 1.5% per month will be charged
- Billing validation/certification to be made within 7 days from the date of provisional bill submission.
- All the prices mentioned in proposal are in rupees and exclusive of GST. Taxes as per applicable
- Minimum contract period is for 3 Years post Go-Live.

6 Previous Experience

1. **A global leader in construction materials manufacturing faced significant challenges in ensuring vehicle safety compliance during loading operations at its production facilities. The company operates in over 50 countries and is known for producing cement, aggregates, and ready-mix concrete.**

Project Scope	• Automate safety compliance checks for vehicles and drivers. • Reduce manual intervention and check time. • Integrate with Vaahan, Sarathi, and SAP for data flow. • Store historical data and images for future audits.
Business Challenges	• Manual Safety Checks: 25 minutes per vehicle. • Human Errors: Risk of compliance lapses. • Data Management: No centralized system for historical records. • Driver Adherence: Ensuring compliance with safety equipment and requirements.
Solution Highlights	• Industrial tablets for document verification checkpoints. • Integrated Vaahan, Sarathi, and SAP systems. • Automated real-time validation of documents like RC, insurance, and permits. • Stored historical data and images in a centralized system. • Manual intervention only for non-validated documents.
Benefits Delivered	• **Time Efficiency:** Reduced check time from 25 minutes to 2 minutes. • **Accuracy:** Eliminated manual errors in document verification. • **Cost Savings:** Reduced dependency on manpower. • **Safety:** Enhanced enforcement of safety protocols. • **Scalability:** Integrated with SAP and government databases for future expansion.
Delivery Process	• Conducted requirement analysis with stakeholders. • Designed automated workflows. • Implemented industrial-grade hardware and integrated systems. • Tested and validated the system. • Trained staff and provided system handover. • Monitored and supported post-implementation.
Outcome	Axestrack's solution streamlined safety compliance, minimized manual interventions, and created a scalable framework for Client future automation needs.

2. **A leading conglomerate in the food and agricultural sector specializing in palm oil production faced challenges in maintaining oil quality during transit, ETA management, automated sequencing and optimizing in-plant operations**

Project Scope	• Real-time vehicle movement monitoring through GPS-based tracking for transparency and efficiency. • ETA calculation and sharing to provide unloading staff with real-time updates, enhancing planning and resource allocation. • Automate sequencing and prioritize delayed vehicles to reduce TAT • Maintain unloading order through FIFO system, ensuring smooth operations.
Business Challenges	• Product quality degradation occurred due to delays in transit and unloading, which affected the freshness of palm fruit. • Inefficient in-plant operations arose from challenges in adhering to the vehicle FIFO system for material unloading. • A lack of visibility resulted from inadequate real-time tracking and the absence of timely ETA communication to unloading staff. • Delayed decision-making was caused by manual prioritization of delayed vehicles, leading to inefficiencies in the process.
Solution Highlights	• Continuous monitoring of vehicles was achieved through real-time GPS tracking, ensuring visibility throughout transit. • ETA notifications automatically informed unloading staff of vehicle arrivals based on real-time tracking, enhancing preparedness. • Delay management dynamically prioritized delayed vehicles, reducing the impact on palm oil quality. • Automated sequencing adjusted the unloading order based on vehicle status and delays, streamlining operations.
Benefits Delivered	• **Enhanced Quality:** Maintained palm oil freshness by minimizing delays. • **Operational Efficiency:** Reduced in-plant TAT through automated processes. • **Improved Visibility:** Real-time updates enhanced coordination between transit and in-plant teams. • **Optimized Decision-Making:** Automated prioritization reduced manual intervention and errors.
Delivery Process	• Engaged with client stakeholders to understand pain points & operational goals. • Developed a tailored solution integrating client ERP, GPS tracking & automated sequencing. • Deployed GPS trackers, integrated ETA systems, and set up automated workflows for prioritization. • Ensured solution accuracy and reliability through end-to-end testing. • Trained client staff to use the system effectively and provided post-implementation support.
Outcome	Axestrack's CPL solution significantly improved operational efficiency at client, ensuring the freshness of palm fruit, reducing delays, and automating in-plant processes for long-term scalability and reliability.

3. **A leading cement manufacturer, faced challenges in vehicle tracking, route deviations, detention, and pilferage across their logistics operations. These issues were further compounded by manual document checks, inefficient trip management, and high operational costs.**

Project Scope	• Implementation of Vehicle Tracking System (VTS) for Client. • Real-time trip tracking and route optimization. • Integration with UltraTech's SAP for data automation. • Reduction of manual interventions in logistics. • Detailed analytics on transportation efficiency.
Business Challenges	• Inefficient manual tracking of vehicles. • High Turnaround Time (TAT) per trip. • Frequent en-route halts and route deviations. • Detention at plants and warehouses. • Pilferage of goods during transportation. • Manual verification of documents and safety compliance, increasing workload. • Lack of data integration with SAP for trip details.
Solution Highlights	• Implemented Unified platform tracks 45,000+ trucks with real-time location, ETA, and halt alerts. • Trip data integrates seamlessly with SAP via API for efficient record-keeping. • Advanced analytics optimize transporter, truck, and driver performance. • Streamlined logistics saved costs across 200,000+ trips and multiple zones over 10+ years. • Automated compliance checks reduced TAT, detention, pilferage, and manual efforts.
Benefits Delivered	• **Enhanced Operational Efficiency:** Automated tracking reduced manual interventions, minimized TAT, and improved logistics speed. • **Cost Optimization:** Real-time monitoring reduced pilferage, unnecessary halts, and fuel wastage, saving operational costs. • **Improved Safety and Compliance:** Automated checks ensured adherence to permits, RC, etc. and regulatory compliance. • **Actionable Insights through Analytics:** Comprehensive data analytics provided valuable insights for optimizing performance and decision-making.
Delivery Process	• Assessed client logistics challenges and mapped pain points. • Implemented Axestrack's unified VTS platform with GPS installation on trucks. • Integrated with SAP via API for seamless data flow and automated trip tracking. • Provided actionable insights through historical data and real-time analytics to refine operations continuously.
Outcome	Axestrack's VTS implementation via its Digital Platform significantly reduced client trip's TAT, pilferage, detention, streamlined logistics operations, saving costs across 200,000+ monthly trips while scaling across multiple zones with a 10+ year partnership.

7 Support and Maintenance Process

CPL is supported through monthly, quarterly, and annual updates to maintain optimal performance and meet evolving needs.

Best Practices for Support and Maintenance:

- **Scheduled Updates:**
  - **Monthly:** Bug fixes, performance optimizations, and minor enhancements.
  - **Quarterly:** Feature releases, system audits, and design updates.
  - **Yearly:** Comprehensive performance reviews, architectural upgrades, and long-term scalability planning.

- **Proactive Monitoring:**
  - Real-time monitoring with Datadog and Firebase Crashlytics to detect and address issues.
  - Alerts and logs are analyzed regularly to ensure system health.

- **Continuous Quality Assurance:**
  - Code analysis with SonarQube to uphold code quality and security.
  - Automated testing pipelines to detect issues before deployment.

- **Collaboration and Feedback**
  - Regular client check-ins to align on feature requests and enhancements.
  - Feedback incorporated into the Agile workflow for iterative improvement.

- **Scalability and Performance Management:**
  - Ongoing optimization of microservices for handling increased loads.
  - Database performance tuning and regular indexing with PostgreSQL.

- **Documentation and Knowledge Sharing:**
  - Maintain up-to-date technical documentation for APIs, workflows, and troubleshooting.
  - Provide training sessions or resources for client teams as needed.

- **Incident Management:**
  - Implement a structured escalation process for resolving critical issues swiftly.
  - Root cause analysis for incidents to prevent recurrence.

With this technical robustness and structured methodology, CPL ensures seamless port operations and long-term client satisfaction.

8 Governance Structure

Axestrack has three layered approaches to Governance

Steering committee and value board: This layer will have the executive sponsor for the project with one-to-one mapping with the ABG Trading Sponsors level management. The team will meet monthly/quarterly to validate and monitor the project progress and take strategic decisions. This layer is the highest level of escalation,

Enagament & project management: This layer provides the project level monitoring and management. They will deal with the day-to-day operation issues and escalations. This layer will be responsible for the project success and excellence.

Project operations & transformation: This layer involves all the members of execution team including Technical SPOCs, functional SPOCs and SMEs. These will interact with the client project team on daily basis and work for the success of the project.

9 Training Approach

User Manual

- Provides in-depth information on solution features and functionalities.
- Includes troubleshooting steps and FAQS.
- Easy-to-navigate sections for quick reference.

Online Assistance

- Tailored sessions addressing specific needs of key stakeholders.
- Real-time resolution of queries and Issues.
- Enhanced understanding through direct interaction.
- Interactive sessions with live Q&A.

Quarterly Training Sessions

- Organized every quarter to keep users informed.
- Detailed walkthroughs of new features and updates.
- Ensures continuous improvement and knowledge enhancement.
- Ensuring team is always at the forefront of industry standards

'Train the Trainer' Approach

- • Designated trainers within organization receive specialized training.
- Efficient knowledge dissemination to all team members.
- Ensuring users are up-to-date (Productive & proficient)
- Regular feedback collection to improve the product.

Train the Trainer model

Who will be trained?

Axestrack will conduct training for select users from each user group and provide them hands on training on the application

How?

- Axestrack will conduct regular trainings as per the business requirement.
- Axestrack will provide FAQ guides and user manuals to assist in trainings

What will be they trained on?

Axestrack will provide training end to end training for the product workflow and solution approach.

Expectations from Client

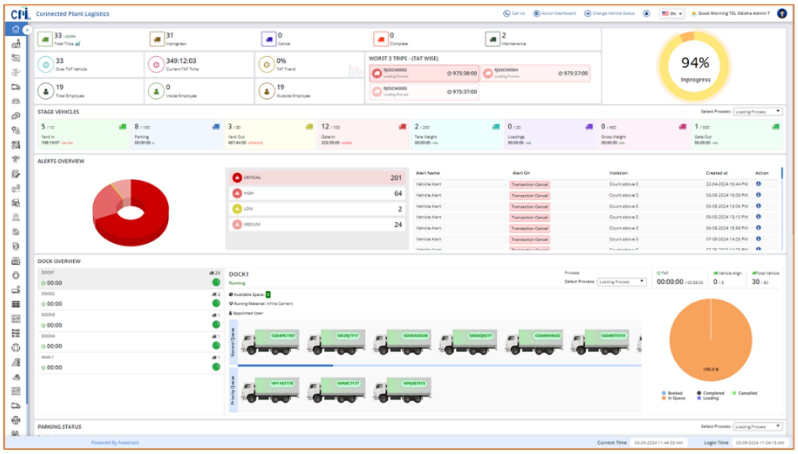
- Nominate the right set of individuals who are enthusiastic and can take the learning to a boarder group
- Detailed stakeholder matrix

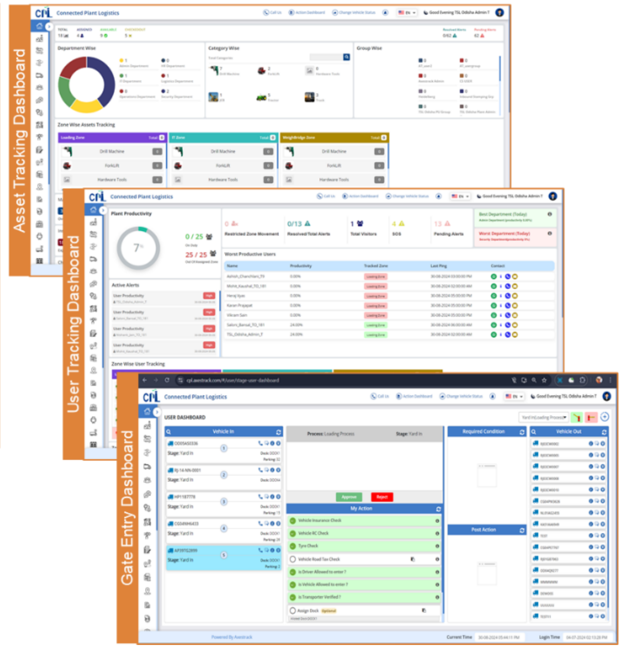
10 Key Assumptions & Dependencies

- Civil work, Mounting poles, Power supply, LAN connectivity, hardware setup, if any, are out of scope for the project
- ABG Trading will provide the timely approvals and sign-off for the work products (Digital Layer) delivered as part of project.
- Coordination with CHA/Third Party Agent for data inputs in CPL digital layer will be done by ABG Trading team.
- Any new requirements beyond the scope of this proposal will be considered as change request for which separate effort & cost to be estimated.
- Data extraction, cleansing, validation & reconciliation from SAP will be done by ABG Trading business users.
- Issues and bugs related to 3rd Party product or services or plugins will be out of scope for this project.
- All project deliverables will be in English.
- Local language support or any translation requirement to be supported by ABG Trading.
- The Network, security, IT and other infrastructure for this project to be owned by ABG Trading team. Axestrack's responsibility is only to provide recommendation related to these areas.
- It will be SaaS deployment solution. On-Premise network configuration will be out of scope.
- Any hardware requirement will be charged separately.
- All communication with SAP will be through API Integration
- Checkpoints/Stages data inputs & validation will be done by ground operation team (ABG Trading / CHA) by dynamic checklist
- ABG Trading team will be providing instruction and sign-off after each testing stage (i.e SIT & UAT).
- Axestrack will be dependent on ABG Trading for any delay due to major deviation, weather/environmental conditions, new requirements and non-availability of ABG Trading SPOCs (users & project team) resulting in project timeline delay will not be Axestrack responsibility.
- Train the trainer approach will be followed for the user training
- ABG Trading need to provide all the required approvals and passes to the Axestrack Onsite team (if required during project implementation) at Port Terminal. Also, ABG Trading will be responsible to provide Safety Training & PPE's to Axestrack's team.

11 Overview of Connected Port Logistics

11.1 One Stop Solution for Port logistics & Assets Visibility

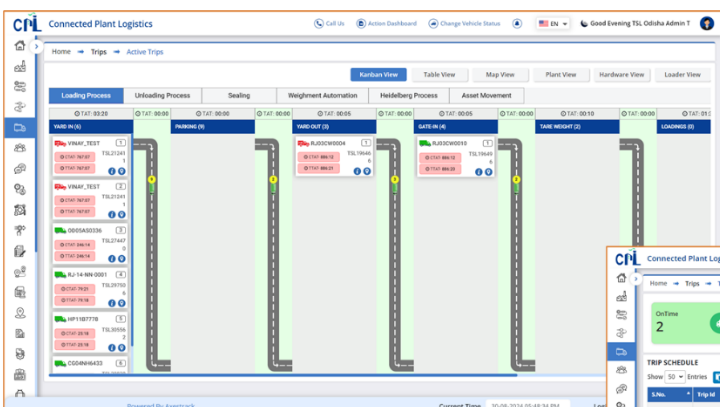




Summary Dashboard

Our dashboards **provides complete control & visibility** across all processes including TEU, Port, Employee & Asset tracking

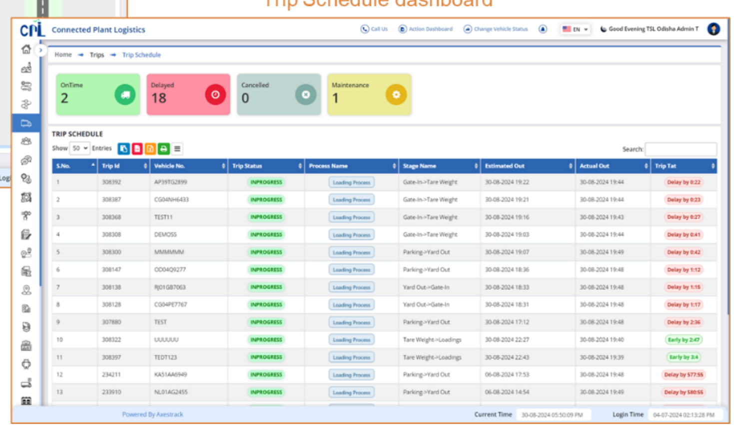
11.2 In-Port Trip Management



Trip Management dashboard

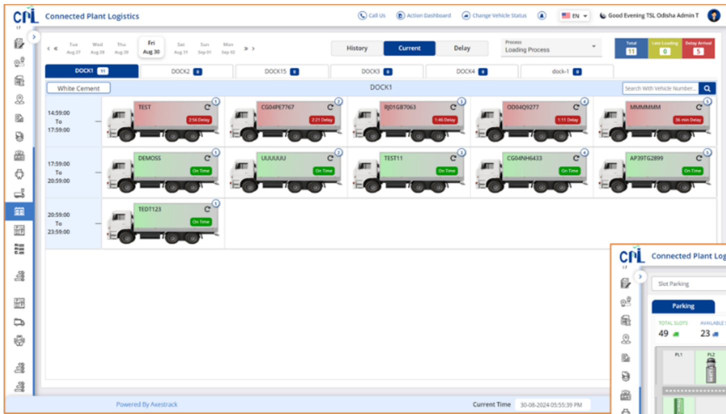
Stage-based tracking to help you measure, analyze and reduce Turnaround Time

Trip Schedule dashboard



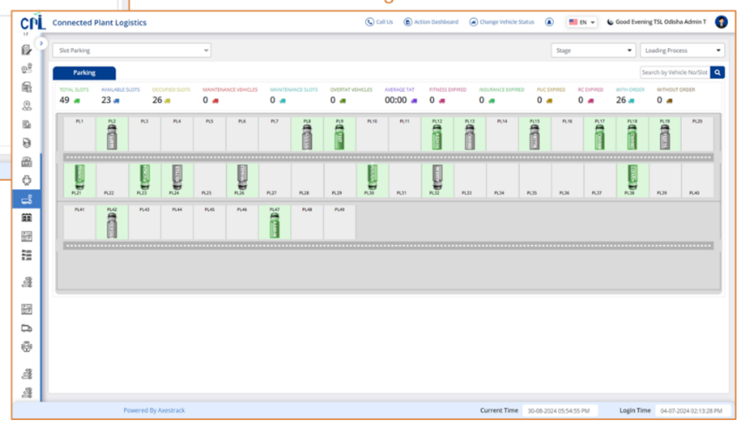
Real Time Trip Status for actively validate bottlenecks and take optimization actions to reduce detentions and streamline the processes.

11.3 SMART Dock & Parking Management

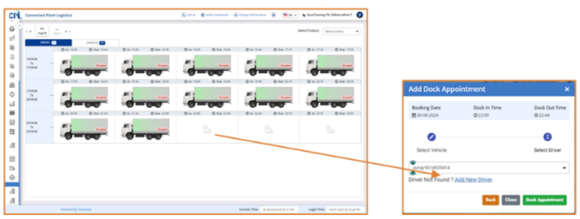


Managing parking and dockyard slots to **reduce idle time** and ensure **smooth vehicle movement** for loading and unloading materials.

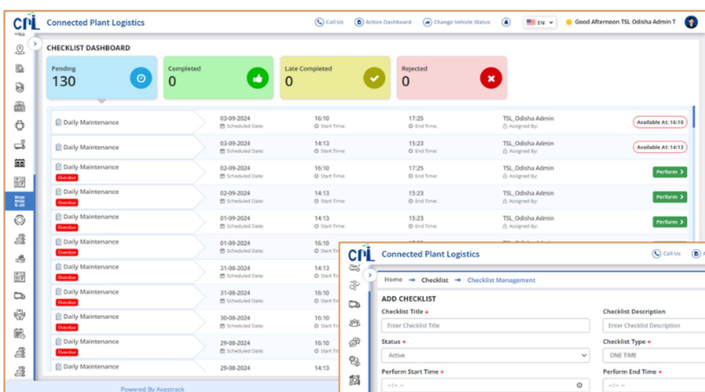
Parking dashboard



Dock Management dashboard

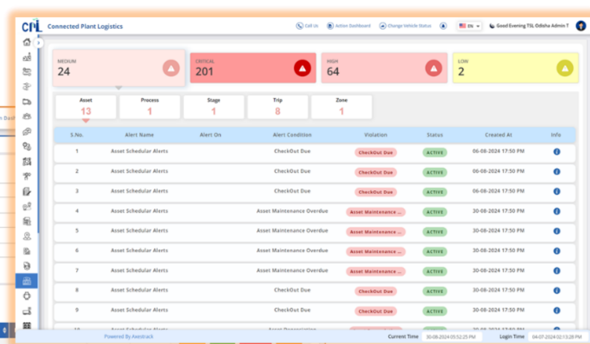


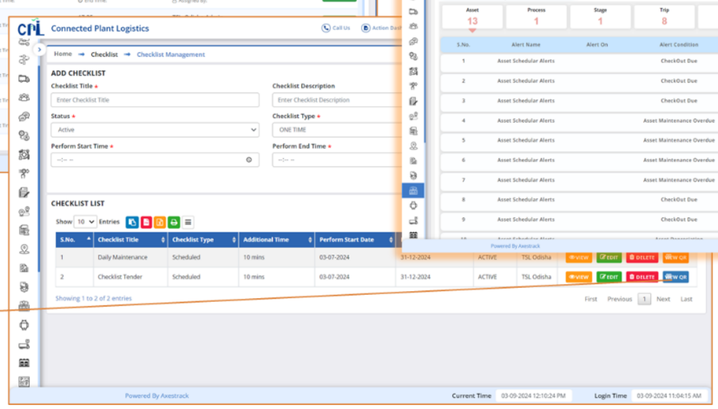
11.4 Checklist & Alerts Dashboard



Optimize Entry/Exit Time and Process for Fleets & proactive Alerts Dashboard for any violations

Alert dashboard





Checklist dashboard

11.5 Benefits of Axestrack Connected Port Logistics

Higher TAT efficiency
Stage wise vehicle management and TAT monitoring at Process, Stage, Between Stage, Destination and Staff Level

Connected Port Logistics

Improved Productivity
Improve Staff efficiency and reduce TAT at checkpoints in the port operations

Real time visibility
Realtime validation from to reduce administrative burden and ensuring regulatory compliance

Improved Safety, Compliance, Smart Auditing
Smart Audits, Optimized Operations, Goods Validation, Cost Effective & Transparent

Real time alerts, notifications, exceptions management
Multi channel notification mechanism to deal with exceptions on real time basis

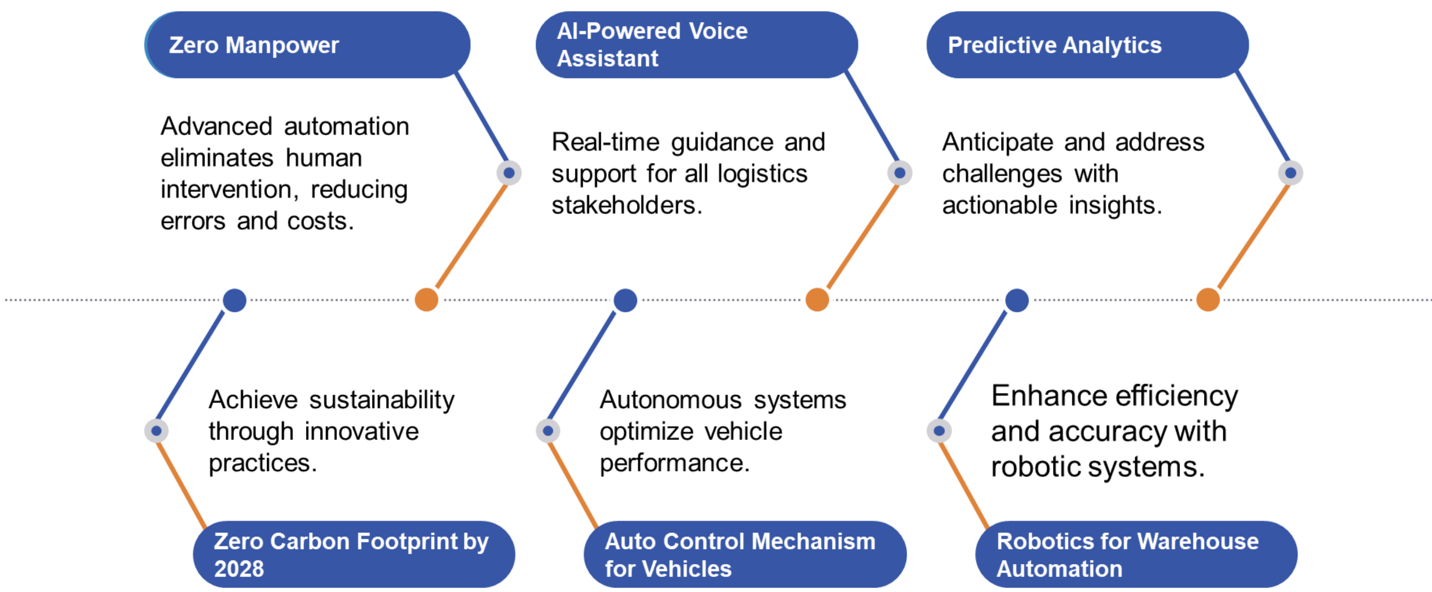
Data to make long term decisions
End-to-end visibility of the Goods (from vessel Discharge to point of Sale) and Placement Co-ordination between logistics team, security team, and other stakeholders.

12 Pragmatic View of Product Roadmap & Innovations Planned

Axestrack’s product roadmap has been built keeping current and future needs of the industry in scope. The roadmap establishes our commitment to the innovations and capabilities which we co-develop with client to ensure these developments are relevant to the needs of the clients.

The intent behind this vision is simple. We understand that “connectivity” is the way for future and hence the core of any solution must embrace the connected part and should support the mechanisms which promotes connectivity.

We understand that while vision is clear, we also need to keep the structural elements flexible and modular. Hence, the design of our architecture and roadmap has been envisioned to be ‘modular and agile’.



AXESTRACK

Right Information,Great Decisions